

# Real-Time Clock with Temperature Logger

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Presented here is a simple real-time clock with temperature logger. The project shows time, date and real-time temperature on the LCD. The logger records the maximum and minimum temperature for each day and can keep entries for 51 days. The recorded temperatures can be viewed any time together with the dates. The temperature is sensed via single-wire temperature sensor and the maximum and minimum temperatures for each day are stored in EEPROM via PC protocol. Fig. 1 shows the author's prototype.

## Circuit and working

Fig. 2 shows the circuit diagram of real-time clock with temperature logger. The circuit is built around microcontroller PIC18F452 (IC1), serial PC real-time clock DS1307 (IC2), EEPROM AT24C02 (IC3), single-wire temperature sensor DS18B20 (SENSOR1) and 16x2 LCD.

Transformer X1 is used to power the circuit. The secondary of X1 is fed to bridge rectifier BR1 which is a general-purpose 1A bridge rectifier. The rectified output is filtered through capacitor C1 and fed to 5V regulator IC4 to get a regulated 5V supply to power the circuit. LED1 is used to indicate the input power status of the circuit and current through it is limited by resistor R1.

**Single-wire temperature sensor.** The real-time temperature is sensed by single-wire digital thermometer DS18B20. The sensor provides 9-bit to 12-bit Celsius temperature measurements and has an alarm function with nonvolatile user-programmable upper and lower trigger points. DS18B20 communicates over a single-wire bus that by definition requires only one data line (and ground) for communication with a central microprocessor. It has an

operating temperature range of -55°C to +125°C and is accurate to  $\pm 0.5^\circ\text{C}$  over the range of  $-10^\circ\text{C}$  to  $+85^\circ\text{C}$ . In addition, the DS18B20 can derive power directly from the data line ('parasite power'), eliminating the need for an external power supply. But we have powered it

AM/PM indicator. The DS1307 has a built-in power sense circuit which detects power failures and automatically switches to the battery supply. A 32.768kHz crystal oscillator provides clock to DS1307 and a 3V battery connected at pin 3 provides the backup

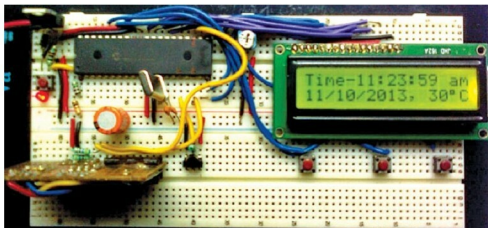


Fig. 1: Author's prototype

externally for this project. The DQ pin 2 of the sensor DS18B20 is interfaced to port pin RC1 of microcontroller IC1 using single-wire protocol through a pull-up resistor R2. Single-wire protocol was developed by Maxim Integrated; it needs only one wire for communication and we can connect many devices with a single wire and access each of them separately according to our need.

**Real-time clock.** Serial PC-based real-time clock DS1307 provides the current date and real time to be displayed on the LCD. DS1307 is a low-power, full BCD clock/calendar with 56 bytes of NV SRAM. Address and data are transferred serially through PC bi-directional bus. The clock/calendar provides seconds, minutes, hours, day, date, month and year information. The end of the month date is automatically adjusted for months with less than 31 days, including corrections for leap year. The clock operates in either the 24-hour or 12-hour format with



## PARTS LIST

### Semiconductors:

- IC1 - PIC18F452, microcontroller
- IC2 - DS1307, RTC
- IC3 - AT24C02, EEPROM
- IC4 - 7805, 5V regulator
- SENSOR1 - DS18B20, 1-wire digital thermometer
- LED1 - 5mm LED
- LCD1 - 16x2 LCD
- BR1 - Bridge rectifier module, 1A

### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 680-ohm
- R2, R7-R9 - 4.7-kilo-ohm
- R3 - 10-kilo-ohm
- R4, R5 - 2.2-kilo-ohm
- R6 - 100-ohm
- VR1 - 10-kilo-ohm preset

### Capacitors:

- C1 - 470 $\mu\text{F}$ , 25V electrolytic
- C2 - 0.1 $\mu\text{F}$  ceramic
- C3 - 10 $\mu\text{F}$ , 16V electrolytic
- C4, C5 - 22pF ceramic

### Miscellaneous:

- X1 - 230V AC primary to 12V, 500mA secondary transformer
- S1-S4 - Tactile switch
- X<sub>OSC</sub>1 - 10MHz crystal oscillator
- X<sub>OSC</sub>2 - 32.768kHz crystal oscillator
- BATT.1 - 3V battery
- CON1 - 2-pin terminal block connector